EXP.NO: 5	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. Input:

- The user fills out a web form with their **name** and **email**.
- The form will use both **GET** and **POST** methods for data submission.

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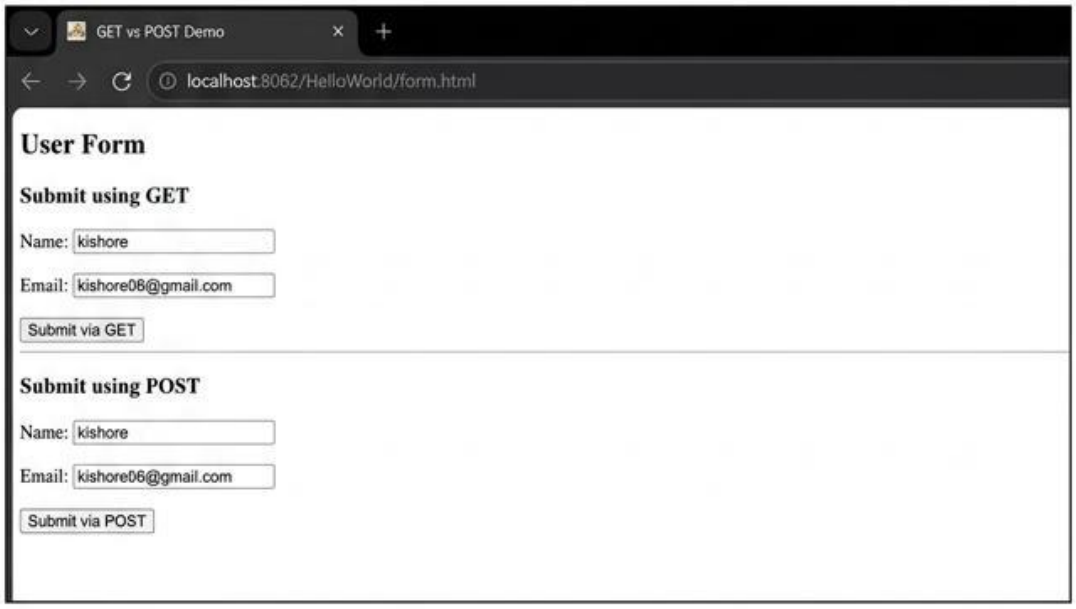
2. Steps for Creating and Handling the HTTP GET and POST Requests:

1. **Create an HTML Form:**
    - Design an HTML form with two input fields: one for the **name** and one for the **email**.
    - Create two buttons to allow the user to submit the form using **GET** or **POST** methods.
    - Use the action attribute to define the **Servlet URL** and specify the form method (either GET or POST).
2. **Create the Servlet to Handle GET and POST Requests:**
    - Write a Java Servlet class that will handle **both GET and POST requests**.
    - The servlet will have two methods:
        - doGet() to handle the GET requests.
        - doPost() to handle the POST requests.
    - In both methods, retrieve the form data using request.getParameter().
    - In the doGet() method, process and display the data in the URL.
    - In the doPost() method, process and display the data in the HTTP request body.

3. **Handle GET Requests in the Servlet:**
    - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
    - Output the data as a response, including the **name** and **email** in the URL (visible to the user).
4. **Handle POST Requests in the Servlet:**
    - In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
    - Output the data as a response, displaying the **name** and **email** without exposing them in the URL (more secure).
5. **Return the Output:**
    - The Servlet should return an HTML page displaying the **name** and **email** entered by the user.
    - For the **GET method**, the output should be visible in the URL.
    - For the **POST method**, the output should be visible in the response body, not the URL.
6. **Deploy and Test the Servlet:**
    - Compile the Servlet class and deploy it to a **Servlet container** (like **Apache Tomcat**).
    - Test the form using **GET** and **POST** methods to see the difference in data submission.
7. **Observe and Compare GET and POST Results:**
    - Test the **GET method**: The form data is sent in the URL.
    - Test the **POST method**: The form data is sent in the HTTP request body, not visible in the URL.

Output:



HTML File (form.html):

```html
<!DOCTYPE html>
<html>
<head>
    <title>GET vs POST Demo</title>
</head>
<body>
    <h2>User Form</h2>

    <!-- GET Form -->
    <form action="GetPostServlet" method="get">
        <h3>Submit using GET</h3>
        Name: <input type="text" name="name"><br><br>
        Email: <input type="email" name="email"><br><br>
        <input type="submit" value="Submit via GET">
    </form>

    <hr>

    <!-- POST Form -->
    <form action="GetPostServlet" method="post">
        <h3>Submit using POST</h3>
        Name: <input type="text" name="name"><br><br>
        Email: <input type="email" name="email"><br><br>
        <input type="submit" value="Submit via POST">
    </form>

</body>
```

```
</html>
```

Java File (GetPostServlet.java)

```
import java.io.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;

public class GetPostServlet extends HttpServlet {

    // Handle GET request
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String email = request.getParameter("email");

        out.println("<html><body>");
        out.println("<h2>GET Method Used</h2>");
        out.println("<p><b>Name:</b> " + name + "</p>");
        out.println("<p><b>Email:</b> " + email + "</p>");
        out.println("<p>Data is visible in the URL</p>");
        out.println("</body></html>");
    }
```

```
    // Handle POST request
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("name");
        String email = request.getParameter("email");

        out.println("<html><body>");
        out.println("<h2>POST Method Used</h2>");
        out.println("<p><b>Name:</b> " + name + "</p>");
        out.println("<p><b>Email:</b> " + email + "</p>");
        out.println("<p>Data is sent in HTTP body (not visible in URL)</p>");
        out.println("</body></html>");
    }
}
```

File (web.xml):

```
<web-app xmlns="https://jakarta.ee/xml/ns/jakartaee"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="https://jakarta.ee/xml/ns/jakartaee
        https://jakarta.ee/xml/ns/jakartaee/web-app_6_0.xsd"
        version="6.0">
```

```
    <servlet>
        <servlet-name>GetPostServlet</servlet-name>
        <servlet-class>GetPostServlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>GetPostServlet</servlet-name>
        <url-pattern>/GetPostServlet</url-pattern>
    </servlet-mapping>
</web-app>
```

Java Compiled File (GetPostServlet.class) :

```
// Source code is decompiled from a .class file using FernFlower decompiler (from Intellij IDEA).
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;

public class GetPostServlet extends HttpServlet {
    public GetPostServlet() {
    }

    protected void doGet(HttpServletRequest var1, HttpServletResponse var2) throws ServletException, IOException {
        var2.setContentType("text/html");
        PrintWriter var3 = var2.getWriter();
```

```
    String var4 = var1.getParameter("name");
    String var5 = var1.getParameter("email");
    var3.println("<html><body>");
    var3.println("<h2>GET Method Used</h2>");
    var3.println("<p><b>Name:</b> " + var4 + "</p>");
    var3.println("<p><b>Email:</b> " + var5 + "</p>");
    var3.println("<p>Data is visible in the URL</p>");
    var3.println("</body></html>");
  }

  protected void doPost(HttpServletRequest var1, HttpServletResponse var2) throws
ServletException, IOException {
    var2.setContentType("text/html");
    PrintWriter var3 = var2.getWriter();
    String var4 = var1.getParameter("name");
    String var5 = var1.getParameter("email");
    var3.println("<html><body>");
    var3.println("<h2>POST Method Used</h2>");
    var3.println("<p><b>Name:</b> " + var4 + "</p>");
    var3.println("<p><b>Email:</b> " + var5 + "</p>");
    var3.println("<p>Data is sent in HTTP body (not visible in URL)</p>");
    var3.println("</body></html>");
  }
}
```

RESULT:

Thus, A Servlet was created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.